Algebra-IV Notes

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1. Modules

Throughout this section R will denote a commutative ring and M will denote an R-module.

Definition 1.1. A left R-module is an abelian group M along with a left-action of R on M given by the ring homomorphism $\sigma : R \to \mathrm{End}_{Ab}(M)$.

Recall that $\mathrm{End}_{Ab}(M)$ is a ring where addition is pointwise and multiplication is composition. The action of $r \in R$ on $m \in M$ is $\sigma(r)(m)$ and is denoted by rm.

Right R-modules are defined analogously with a right action instead. Generally, every right R-module structure is associated with a left R°-module structure where R° denotes the opposite ring (order of multiplication in R is reversed). Hence, if R is commutative, the left and right R-module structures coincide.

Definition 1.2. A subgroup $N \leqslant M$ is a submodule of M if it is closed under action of R, i.e. $RN \subseteq N$.

Let $S \subset M$, we denote by (S) the submodule generated by S, i.e. the intersection of all submodules that contain S. Alternatively, it is the set of all finite R linear combination of elements of S.

Definition 1.3. Let M be an R-module. An element $x \in M$ is called a *torsion element* if there exists $a \in R \setminus \{0\}$ such that $a \cdot x = 0$. We denote by M_{tor} the submodule consisting of all torsion elements of M. If $M_{tor} = M$ then M is called a *torsion module* and if $M_{tor} = \{0\}$ then M is called *torsion free*.

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We now present some general results about Annihilators.

Definition 1.4. We define the *Annihilator* of $A \subset M$, denoted $\mathrm{Ann}_R(A)$, to be

$$\mathrm{Ann}_R(A) = \{r \in R \mid r \cdot a = 0 \ \forall a \in A\}$$

Observe that $\mathrm{Ann}_R(A)$ is a ideal of R.

Definition 1.5. Let $M = M_p$ for some prime $p \in R$. Let $x \in M \setminus \{0\}$. The *exponent* of x is the integer e such that $p^e x = 0$ and $p^{e-1}x = 0$.

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If M is finitely generated, then $\mathrm{Ann}_R(M)$ is nonzero. Indeed, let $M = (x_1, \dots, x_n)$ then $\mathrm{Ann}_R(Rx_i) \neq 0$. Observe that

$$\bigcap_{i=1}^{n} \mathrm{Ann}_R(Rx_i) = \mathrm{Ann}_R(M).$$

If $c_i \in \mathrm{Ann}_R(Rx_i)$ for each i, then $c_1 \cdots c_n \in \mathrm{Ann}_R(M)$.

Definition 1.6. Let $p \in R$ be a prime. We define

$$M(p) = \{x \in M \mid \exists e > 0 \text{ such that } p^e x = 0\}.$$

If R is Noetherian, then there is an $n > 0$ such that p^n kills $M(p)$. Indeed,

$$\{x \in M \mid px = 0\} \subset \{x \in M \mid p^2 x = 0\} \subset \cdots$$

is an ascending chain, and since R is Noetherian, this chain stabilizes.

Further, let $p, q \in R$ be distinct primes, then $M(p) \cap M(q) = 0$. Indeed, if $x \in M(p) \cap M(q)$ then there are r, s such that $p^r x = 0 = q^s x$. Since $(p, q) = 1$, then $(p^r, q^s) = 1$ and hence there are $a, b \in R$ such that $ap^r + bq^s = 1$. Multiplying throughout by x,

$$x = ap^r x + bq^s x = 0.$$

1.1. Homomorphisms

Definition 1.7. Let M, N be R-modules. A homomorphism of groups $\varphi : M \to N$ is a module homomorphism if it respects the action of R, that is

$$\varphi(r \cdot m) = r \cdot \varphi(m) \qquad (\forall r \in R)\ (\forall m \in M).$$

Remark 1.8. Let M and N be R-modules. The abelian group $\mathrm{Hom}_{R\text{-Mod}}(M, N)$ can be made into an R-module by defining $(r \cdot \varphi)(x) = r \cdot \varphi(x)$ for all $x \in M, r \in R$.

Let R and S be commutative rings and let $\alpha : R \to S$ be a ring homomorphism. If M is an R-module then we can use this homomorphism to make M into an S-module by defining $s \cdot x = \alpha(s) \cdot x$ for all $x \in M$. Further, if $\alpha(M)$ is contained in the center of S then α is called an algebra over R.

1.2. Quotient Modules

Definition 1.9. If N is a submodule of M then the quotient M/N as abelian groups can be given an R-module structure by defining

$$r \cdot \overline{m} = \overline{rm} \qquad\qquad (\forall r \in R)\ (\forall \overline{m} \in M/N).$$

Remark 1.10. Let $I \subset R$ be an ideal. We define

$$IM = \left\{\sum i_j m_j \mid i_j \in I, m_j \in M, j \in \mathbb{N}\right\}.$$

Observe that $IM \subset M$ is a submodule. We show that M/IM is an R/I-module. Observe that I annihilates M/IM since

$$i \cdot (m + IM) = im + IM = IM \qquad (\forall\, i \in I)\ (\forall\, m \in M).$$

By the universal property of quotients, we get a homomorphism $\widetilde{\sigma} : R/I \to \mathrm{End}(M/IM)$, proving our claim.

$$\begin{tikzcd}
R \arrow[r, "\sigma"] \arrow[d, "\pi"'] & \mathrm{End}(M/IM) \\
R/I \arrow[ur, "\exists! \widetilde{\sigma}"'] &
\end{tikzcd}$$

More concretely, the action is given by $\pi(r) \cdot \overline{m} = \overline{r \cdot m}$ where $r \in R$ and $m \in M$.

1.3. Products and Coproducts

Definition 1.11. Given R-modules $\{M_i\}_{i\in I}$, we define their direct product $\prod M_i$ as the direct product of abelian groups equipped with the action of R given by $r\cdot(a_i)_{i\in I}=(r\cdot a_i)_{i\in I}$ for all $(a_i)_{i\in I}\in\prod M_i, r\in R$.

Recall that in the (infinite) direct sum of abelian groups $\bigoplus A_i$, the elements are of the form $(a_i)_{i\in I}$ where $a_i\in A_i$ and all but finitely many a_i are 0. Indeed, the direct sum of modules is the coproduct in R-Mod and the finite condition can be deduced from the universal property of coproducts.

We define the direct sum of modules $\bigoplus M_i$ to be the direct sum of abelian groups equipped with the action of R given by $r\cdot(a_i)_{i\in I}=(r\cdot a_i)_{i\in I}$ for all $(a_i)_{i\in I}\in\bigoplus M_i, r\in R$.

Definition 1.12. We call $y_1,\dots,y_n\in M$ independent if whenever there are $a_i\in R$ for which

$$a_1y_1+\cdots+a_ny_n=0$$

gives $a_iy_i=0$ for all i. Note that linear independence and independence are not related. Further if $y_1,\dots,y_n$ are independent then

$$(y_1,\dots,y_n)=(y_1)\oplus\cdots\oplus(y_n).$$

Theorem 1.13. *(Splitting of Exact Sequence) Let $0\to M'\xrightarrow{f}M\xrightarrow{g}M''\to 0$ be exact. Then the following are equivalent:*

(1) There exists a homomorphism $\varphi: M''\to M$ such that $g\circ\varphi=id$.

(2) $M=M'\oplus M''=\ker g\oplus\operatorname{Im}\varphi=\operatorname{Im} f\oplus\ker\psi$.

(3) There exists a homomorphism $\psi: M\to M'$ such that $\psi\circ f=id$.

When these conditions are satisfied, we say ψ splits over f or φ splits over g.

Proof. We first show $(1)\implies(2)$. Assume there exists $\varphi: M''\to M$ such that $g\circ\varphi=id$. Then for $x\in M$, it is easily checked that $x-\varphi g(x)\in\ker g$. Hence $x=(x-\varphi g(x))+\varphi g(x)$ for all $x\in M$, the former is in $\ker g$ and the latter in $\operatorname{Im}\varphi$. Now, suppose $x\in\ker g\cap\operatorname{Im}\varphi$ then there exists $y\in M''$ such that $\varphi(y)=x$. Now, $g\varphi(y)=y$ and $g(x)=0$ hence $y=0$ and $x=\varphi(y)=0$. This shows $M=\ker g\oplus\operatorname{Im}\varphi$. Since the sequence is exact, $\ker g=\operatorname{Im} f\cong M'$. We trivially have $\operatorname{Im}\varphi\cong M''$ hence $M=M'\oplus M''$.

$$\begin{tikzcd}
 & & M'\oplus M'' \arrow[dl, "\pi"'] & & \\
0 \arrow[r] & M' \arrow[r, "f"] & M \arrow[u, "\lambda"] \arrow[r, shift left, "g"] & M'' \arrow[l, shift left, "\varphi"] \arrow[r] & 0
\end{tikzcd}$$

We now show $(2)\implies(3)$. From the above construction, we get an isomorphism $\lambda: M\to M'\oplus M''$ where $x\mapsto(f^{-1}(x-\varphi g(x)), g\varphi g(x))$. Note that we can apply f^{-1} since $x-\varphi g(x)$ is in $\ker g$ and hence in $\operatorname{Im} f$ where f^{-1} is well behaved. Let $\pi: M'\oplus M''\to M'$ be the projection map. Define $\psi=\pi\lambda$, this is a homomorphism since π and λ are homomorphisms. It remains to check that $\psi f=id$. Using the fact that $\operatorname{Im} f=\ker g$, for $x\in M'$ we have

$$\begin{aligned}\psi f(x)&=\pi(\lambda(f(x)))\\&=f^{-1}(f(x)-\varphi g f(x))\\&=f^{-1}f(x)=x.\end{aligned}$$

This proves our claim. We can apply a similar argument to show $(3)\implies(2)\implies(1)$. $\square$

Definition 1.14. A set $S \subset M$ is called linearly independent over R if a linear combination

$$\sum_{x \in S} a_x x = 0 \implies a_x = 0 \qquad (\forall x \in S).$$

If S is linearly independent and if two linear combinations

$$\sum_{x \in S} a_x x \qquad \text{and} \qquad \sum_{x \in S} b_x x$$

are equal then $a_x = b_x$ for all $x \in S$ since $\sum_{x \in S}(a_x - b_x)x = 0$.

Definition 1.15. Let M be an R-module and $S \subset M$. We call S a basis if it generates M and is linearly independent. We call M a free module if it has a basis.

Free modules also satisfy the following universal property. Let N be an R-module. Then for any set function $f : S \to N$, there is a unique homomorphism $\Psi : M \to N$ such that $\Psi(s) = f(s)$ for all $s \in S$. The following diagram commutes.

$$\begin{tikzcd} S \arrow[r, hook] \arrow[dr, "f"'] & M \arrow[d, "\exists!\Psi"] \\ & N \end{tikzcd}$$

Remark 1.16. Let M be an R-module then there exists a free module F which surjects onto M.

Proof. Let $F = \bigoplus_{m \in M} R_m$ where $R_m = R$ for all $m \in M$. Define $f : F \to M$ by mapping $e_m \mapsto m$ for all $m \in M$. By the universal property of free modules, this can be extended to a unique module homomorphism and it is clearly surjective. $\square$

Theorem 1.17. *(Invariance of Rank) Let M be a free R-module. We define the rank of M to be the cardinality of its basis. The rank of a free module is well defined.*

Proof. We denote the image of $m \in M$ in M/IM by $\overline{m}$ and the image of $r \in R$ in R/I by $\pi(r)$.

Let $S \subset M$ be a basis. We show that $\overline{S}$ is a basis for M/IM as an R/I-module (see 1.10). We first check that $\overline{S}$ is a generating set. Observe that for any $m \in M$, there are finitely many $r_i \in R$ and $s_i \in S$ for $i = 1 \ldots n$, such that

$$\overline{m} = \overline{\sum r_i s_i} = \sum \overline{r_i s_i} = \sum \pi(r_i) \cdot \overline{s}_i.$$

Now, suppose there are $r_i \in R/I$ and $s_i \in S$ such that

$$\sum \pi(r_i) \cdot s_i = 0 \implies \overline{\sum r_i s_i} = 0 \implies \sum r_i s_i \in IM.$$

We may now write $\sum r_i s_i = \sum i'_j m_j$ for some $i'_j \in I$ and $m_j \in M$. Since S generates M, we may express m_j as an R-linear combination of S. Upon simplification of the RHS, we get $\sum i'_j m_j = \sum i_k s_k$ for some $i_k \in I$. Reordering if necessary, we get $\sum r_k s_k = \sum i_k s_k$. Since S is a basis, we get $r_k = i_k \in I$ for all k. Hence $\pi(r_k) = 0$ in R/I, showing that S is linearly independent.

Observe that the linear independence of $\overline{S}$ shows that $S \leftrightarrow \overline{S}$. If we take I to be a maximal ideal, then R/I is a field and hence M/IM is a vector space over R/I. If M has two bases S and S', then $S \leftrightarrow \overline{S} \leftrightarrow \overline{S'} \leftrightarrow S'$, where the second bijection is due to bases having the same cardinality in vector spaces. $\square$

We now define projective modules.

Definition 1.18. Let P be an R-module. The following are equivalent and define what is called a projective module.

(1) Given a homomorphism $f : P \to M'$ and a surjective homomorphism $g : M \to M'$, then there exists homomorphism $h : P \to M$ such that the following diagram commutes.

$$\begin{tikzcd}
& P \arrow[dl, "\exists h"'] \arrow[d, "f"] & \\
M \arrow[r, "g"] & M' \arrow[r] & 0
\end{tikzcd}$$

(2) Every exact sequence $0 \to M \to M' \to P \to 0$ splits.

(3) There is a module M such that $P \oplus M$ is free, in other words, P is a direct summand of a free module.

Proof. Assume (1). We get the following commutative diagram, showing that the exact sequence splits.

$$\begin{tikzcd}
& P \arrow[dl, "\exists h"'] & \\
M' \arrow[r] & P \arrow[u, "id"] \arrow[r] & 0
\end{tikzcd}$$

Assume (2). By 1.16, there is a free module F such that $F \xrightarrow{\varphi} P \to 0$ is exact. From our assumption this splits and hence $F = P \oplus \ker \varphi$ by 1.13.

Assume (3). Let $f_1 : P \to M''$ and $f_2 : M \to M''$ be homomorphisms. Then by the universal property of direct sums, there is a unique f such that $i_1 f = f_1$ and $i_2 f = f_2$ where i_1, i_2 are the canonical injections. Since $P \oplus M$ is free, it is projective and hence there is an $h : P \circ M \to M'$ such that $gh = f$. Let $h_1 = hi_1$, this shows that P satisfies (1).

$$\begin{tikzcd}
P \arrow[dr, hook, "i_1"] \arrow[dd, "h_1"'] \arrow[ddr, bend right, "f_1"] & & M \arrow[dl, hook', "i_2"'] \arrow[ddl, bend left, "f_2"] \\
& P \oplus M \arrow[d, "f"] \arrow[dl, "h"'] & \\
M' \arrow[r, "g"'] & M'' \arrow[r] & 0
\end{tikzcd}$$

Further note that a free module is projective. $\square$

1.5. Noetherian Modules

Definition 1.19. Let M be an R-module. The following are equivalent and define a *Noetherian Module*.

(1) M satisfies the ascending chain condition.

(2) Every nonempty collections of submodules contains a maximal element.

(3) Every submodule of M is finitely generated.

Proof. Assume (1). Suppose, for the sake of contradiction, that there is a collection $\mathfrak{C}$ of submodules of M that do not contain a maximal element. Since it is nonempty, there is some $N_1 \in \mathfrak{C}$. Having picked $N_1, \ldots, N_k$; since N_k is not maximal, there is some $N_{k+1} \in \mathfrak{C}$ such that $N_k \subsetneq N_{k+1}$. Observe that $\{N_k\}$ is an infinite ascending chain.

Assume (2). Let $N \subset M$ be a submodule. Let $\mathfrak{C}$ be the collection of finitely generated submodules of N. Since $0 \in \mathfrak{C}$, it is nonempty and hence there exists a maximal element $N_1 \in \mathfrak{C}$ by our assumption. We claim that $N_1 = N$. Suppose not, then there is some $n \in N \cap N_1^c$. Observe that $N_1 \subsetneq N_1 + (n)$, contradicting its maximality.

Assume (3). Let $N_1 \subseteq N_2 \subseteq \cdots \subseteq N_k \subseteq \cdots$ be an ascending chain of submodules in M. Observe that $N = \cup_{i=1}^{\infty} N_i$ is a submodule, hence $N = (x_1, \ldots, x_n)$ by our assumption. Then there are $x_k \in N_{i_k}$ for some i_k and $1 \leqslant k \leqslant n$. Let $m = \max(i_1, \ldots, i_n)$. For $j \geqslant m$ we have,

$$N = \sum_{i=1}^{n} Rx_i \subseteq N_j \subseteq N.$$

Hence $N = N_j$ for all $j \geqslant m$. □

Theorem 1.20. *Let M, N be R-modules. If $\varphi : M \to N$ is a surjective ring homomorphism then N is Noetherian.*

Proof. Let $K \subset N$ be a submodule. Since $K' = \varphi^{-1}(K)$ is a submodule of M, it is finitely generated. Let $K' = (x_1, \ldots, x_n)$ for some $x_i \in M$. We claim that $K = (\varphi(x_1), \ldots, \varphi(x_n))$. Suppose $y \in K$ and let $y' \in \varphi^{-1}(y)$, then there are $r_1, \ldots, r_n \in R$ such that

$$r_1x_1 + \cdots + r_nx_n = y' \implies r_1\varphi(x_1) + \cdots r_n\varphi(x_n) = y.$$

Now for the reverse inclusion, for any $r_1, \ldots, r_n \in R$,

$$r_1\varphi(x_1) + \cdots + r_n\varphi(x_n) = \varphi(r_1x_1 + \cdots + r_nx_n) \in K.$$

This shows that all submodules of N are finitely generated, hence N is Noetherian. □

Theorem 1.21. *Let M be an R-module and $N \subseteq M$ be a submodule. Then M is Noetherian if and only if N and M/N are Noetherian.*

Proof. Suppose M is Noetherian. Since all submodules of N are also submodules of M, they are finitely generated, showing that N is Noetherian. Let $\pi : M \to M/N$ be the canonical projection. It follows from 1.20 that M/N is Noetherian.

We now prove the backward direction. Let P be a submodule of M. Since $P \cap N$ is a submodule of N, $P \cap N$ is finitely generated. By the Second Isomorphism theorem,

$$\frac{P}{P \cap N} \cong \frac{P + N}{N}.$$

Hence $P/(P \cap N)$ is finitely generated. It follows that P is finitely generated. □

Remark 1.22. It follows from the preceeding proposition that if R is a Noetherian ring and M is a finitely generated R-module then M is Noetherian. Indeed, then there will be a surjective homomorphism $R^n \to M$. It suffices to show that R^n is Noetherian. We proceed by induction on n. It is trivially true for $n = 1$, assume the result for $n > 1$ then $R^{n+1}/R^n \cong R$.

Corollary 1.23. *Let R be a Noetherian ring and let M be a finitely generated R-module, then M is Noetherian.*

Proof. Since M is finitely generated, there is a surjective map $\varphi : R^n \to M$. By 1.22, we get R^n is Noetherian. Our proposition follows from 1.20. □

Theorem 1.24. *If M is Noetherian then any surjective endomorphism is an isomorphism.*

Proof. Let $f : M \to M$ be surjective. We get the ascending chain of submodules

$$\ker f \subset \ker f^2 \subset \cdots \subset \ker f^n \subset \cdots .$$

Since M is Noetherian, there is n such that $\ker f^n = \ker f^m$ for all $m \geqslant n$. We claim that $\ker f^n \cap \operatorname{Im} f^n = \{0\}$. Indeed, if $x \in \ker f^n \cap \operatorname{Im} f^n$ there exists $y \in M$ such that $f^n(y) = x$ and $f^n(x) = 0$. Then $f^{2n}(y) = 0$, but $\ker f^{2n} = \ker f^n$, hence $0 = f^n(y) = x$. Further, since f is surjective, $\operatorname{Im}(f^n) = M$. Hence $\ker f^n = 0 \implies \ker f = 0$, showing f is injective. □

Theorem 1.25. *(Hilbert) If R is Noetherian then $R[x]$ is Noetherian.*

Proof. Assume, for the sake of contradiction, that there is an ideal $I \subset R[x]$ that is not finitely generated. Since $I \neq 0$, let $f_1 \in I \setminus \{0\}$ be of the smallest possible degree (by the well ordering of $\mathbb{Z}$). Inductively, pick $f_k \in I \setminus (f_1, \dots, f_{k-1})$ for all $k > 1$ of the smallest possible degree. Let $d_i = \deg f_i$, then $d_1 \leqslant d_2 \leqslant \cdots$; and let b_i be the leading coefficient of f_i. Let $J = (b_1, b_2, \dots)$ be an ideal of R and, since R is Noetherian[1], there is k such that $J = (b_1, \dots, b_k)$. Then there are $r_1, \dots, r_k \in R$ such that $b_{k+1} = \sum_{i=1}^k r_i b_i$. Consider

$$g = \sum_{i=1}^{k} r_i x^{d_{k+1}-d_i} f_i,$$

then $\deg g = d_{k+1}$ and its leading coefficient is b_{k+1}. Also $g \in (f_1, \dots, f_k)$, hence $f_{k+1} - g \notin (f_1, \dots, f_k)$. But $\deg f_{k+1} - g < d_{k+1}$, a contradiction. □

It follows by induction that if $R[x_1, \dots, x_n]$ is Noetherian for all $n \geqslant 1$.

1.6. Modules over PIDs

Henceforth, R shall denote a PID. We work towards proving the structure theorem for finitely generated modules over PIDs.

Theorem 1.26. *Let M be a free R-module. If $N \subset M$ is a submodule, then N is free and* $\operatorname{Rank}(N) \leqslant \operatorname{Rank}(M)$.

Proof. We prove this theorem for the case when M has a finite basis, call it $\{x_1, \dots, x_n\} \subseteq M$. Let $N_r = N \cap (x_1, \dots, x_r)$. We prove the theorem by induction on r. Observe that $N_1 = N \cap (x_1) = (rx_1)$ for some $r \in R$ (maybe 0), hence N_1 is free and $\operatorname{Rank}(N_1) \leqslant 1$. Assume the statement for N_r. Let $\mathfrak{a}$ denote the set of all $a \in R$ such that there exists $x \in N$ and there exist $b_1, \dots, b_r \in R$ such that

$$x = b_1 x_1 + \cdots + b_r x_r + a x_{r+1}.$$

Observe that $\mathfrak{a}$ is an ideal of R, and since R is a PID, there is $a_{r+1} \in R$ such that $\mathfrak{a} = (a_{r+1})$. If $a_{r+1} = 0$ then $N_{r+1} = N_r$ proving our claim. If $a_{r+1} \neq 0$, let $w \in N_{r+1}$ be such that the component of x_{r+1} in w is a_{r+1}. Observe that if $x \in N_{r+1}$ then the component of x_{r+1} in x is divisible by a_{r+1}, i.e. $\exists c \in R$ such that $x - cw \in N_r$. Moreover, notice that $N_r \cap (w) = 0$. Hence

$$N_{r+1} = N_r \oplus (w),$$

showing that $\operatorname{Rank}(N_{r+1}) = \operatorname{Rank}(N_r) + 1 \leqslant r + 1$. □

Corollary 1.27. *Let E be a finitely generated R-module and $E' \subset E$ be a submodule. Then E' is finitely generated.*

Proof. Since E is finitely generated, we get a surjective map $\varphi : R^n \to E$. Observe that $\varphi^{-1}(E')$ is a submodule of F. It follows from 1.26 that $\operatorname{Rank}(\varphi^{-1}(E')) \leqslant n$. It follows that E' is finitely generated. □

[1]More precicely, from the ascending chain condition.

Theorem 1.28. *Let E be a finitely generated R-module. Then E/E_{tor} is free. Further, there is a free submodule $F \subset E$ such that*
$$E = E_{tor} \oplus F.$$

Proof. We first show that E/E_{tor} is torsion free. Indeed, for $\overline{x} \in E/E_{tor}$ notice that if there is $b \in R \setminus \{0\}$ such that $b \cdot \overline{x} = 0$, then $b \cdot x \in E_{tor}$. Hence there is $c \in R \setminus \{0\}$ such that $cb \cdot x = 0 \implies x \in E_{tor}$. This shows that E/E_{tor} is torsion free. Note that E/E_{tor} is also finitely generated since E is finitely generated.

We now prove two lemmas to aid our proof.

Lemma 1.29. *If M is a torsion free finitely generated module then it is free.*

Proof. Let $X = \{x_1, \dots, x_n\}$ be a finite set of generators of M and let $\{v_1, \dots, v_m\}$ be the maximal linearly independent subset of X by inclusion. For each $x_i \in X$, $i = 1, \dots, n$, there is $a_i, b_1, \dots, b_m \in R$ such that
$$a_i x_i + b_1 v_1 + \cdots + b_m v_m = 0.$$
We must have $a_i \neq 0$, otherwise the maximality of $\{v_1, \dots, v_m\}$ would be contradicted. Hence $a_i x_i \in (v_1, \dots, v_m)$. Let $a = a_1 a_2 \cdots a_n$, then $aM \subset (v_1, \dots, v_m)$. Consider the homomorphism $\varphi : M \to M$ given by $x \mapsto ax$. Observe that φ is injective since M is torsion free. Now $M = \mathrm{Im}(\varphi) \subset (v_1, \dots, v_n)$, a free module. It follows from 1.26 that M is free. □

The above lemma shows that E/E_{tor} is free.

Lemma 1.30. *Let E, E' be modules with E' free and let $f : E \to E'$ be surjective. Then there is a free $F \subset E$ such that $E = F \oplus \ker f$.*

Proof. Let $\{x_i'\}_{i \in I}$ be a basis for E'. For each i, let $x_i \in E$ be such that $f(x_i) = x_i'$. Let $F = (x_i)_{i \in I}$, we show that F is free. Let $a_i \in R$ be such that
$$\textstyle\sum_{i \in I} a_i x_i = 0 \implies f(\sum_{i \in I} a_i x_i) = 0 \implies \sum_{i \in I} a_i x_i' = 0.$$
Since $\{x_i'\}_{i \in I}$ is a basis, we get $a_i = 0$ for all $i \in I$, thereby showing that F is free.

Given $x \in E$, there are elements $a_i \in R$ such that
$$\textstyle f(x) = \sum_{i \in I} a_i x_i',$$
then $x - \sum_{i \in I} a_i x_i \in \ker f$. It is clear that $F \cap \ker f = 0$, hence $E = F \oplus \ker f$. □

Applying this lemma to $E \to E/E_{tor}$, we get that $E = E_{tor} \oplus F$ where $F \cong E/E_{tor}$. □

We present a partial converse to the argument used in 1.17.

Remark 1.31. Let R be a PID, $p \in R$ a prime and $M = M(p)$ an R-module. Let $x_1, \dots, x_n \in M$ be such that $\overline{x}_1, \dots, \overline{x}_n \in M/pM$ generate M/pM as a R/p vector space, then $x_1, \dots, x_n$ generate M as an R-module.

Proof. Let $N = \sum_{i=1}^n R x_i \subset M$, we shall show the reverse inclusion. Let $y \in M$, then there exist $a_1, \dots, a_m \in R$ such that
$$\overline{y} = \sum_{i=1}^n a_i \overline{x}_i \qquad \text{and let} \qquad z = \sum_{i=1}^n a_i x_i.$$
Let y be a representative of $\overline{y}$, then $y - z \in pM$. We can hence say that,
$$(\forall y \in M)\ (\exists z \in N) \text{ st. } y - z \in pM.$$

Now, $y - z = py_1$ for some $y_1 \in M$. Applying the above statement to y_1, we get that $\exists z_1 \in N$ such that $y_1 - z_1 \in pM$, i.e. $y - z - pz_1 \in p^2M$. Again, we write $y - z - pz_1 = p^2y_2$ for some $y_2 \in M$ and continue this procedure. Since there is some power of p that annihilates M, say e, we get that

$$y - \overbrace{z - pz_1 - p^2z_2 - \cdots p^e z_e}^{\in N} = \cancel{p^{e+1}M} \to 0$$

Hence $y \in N$, proving our proposition. □

Corollary 1.32. *Every minimal generating set (by inclusion) of M has the same cardinality, viz.* $\dim_{R/p} M/pM$.

Proof. Let $\{x_1, \ldots, x_m\}$ be a minimal generating set. Then $\{\overline{x}_1, \ldots, \overline{x}_m\}$ generates M/pM by 1.17. Let $d = \dim_{R/p} M/pM$, then $m \geqslant d$. Suppose, for the sake of contradiction, that $m > d$. We can assume, without loss of generalty, that $\{x_1, \ldots, x_d\}$ generate M/pM as R/p-vector space. Then by 1.31, $\{x_1, \ldots, x_d\}$ generates M, contradicting the minimality of $\{x_1, \ldots, x_m\}$. □

Theorem 1.33. *Let E be a finitely generated torsion module. Let $\mathrm{Ann}_R(M) = (a)$ for some $a \in R$ and let $a = p_1^{\alpha_1} \cdots p_n^{\alpha_n}$ be the prime factorisation of a in R. Then*

$$E = \bigoplus_{i=1}^{n} E(p_i).$$

Proof. It is enough to show that if $r \in R$ can be expressed as $r = bc$ with $(b, c) = 1$, then $E_r = E_b \oplus E_c$. Our result follows by induction. Since $(b, c) = 1$, there are $x, y \in R$ such that

$$bx + cy = 1.$$

Let $s \in E_r$, then

$$bxs + cys = s.$$

Observe that $bxs \in E_c$ since $cbxs = xas = 0$, similarly $cys \in E_b$. Further observe that $E_b \cap E_c = 0$ since if $s \in E_b \cap E_c$ then $bxs + cys = s \implies s = 0$. Hence $E_r = E_b \oplus E_c$.

Use the fact that $E(p) = E_{p^{\alpha_1}}$. □

Remark 1.34. Let E be a torsion module with $\mathrm{Ann}_R(E) = p^r$ for some $r \geqslant 1$ for some prime $p \in R$. Let $x_1 \in E$ have exponent r. Let $\overline{E} = E/(x_1)$ and let $\overline{y}_1, \ldots, \overline{y}_n \in \overline{E}$ be independent elements. Then there are lifts $y_i \in E$ of each $\overline{y}_i$ for each i, such that the exponent of y_i is the same as that of $\overline{y}_i$. Further, $x_1, y_1, \ldots, y_n$ are independent.

Proof. Let $\overline{y} \in \overline{E}$ have exponent n for some $n \geqslant 1$. Let $y \in E$ be a lift of $\overline{y}$, then $p^n y \in (x_1)$, hence

$$p^n y = p^s c x_1 \qquad c \in R, p \nmid c$$

for some $s \leqslant r$. Observe that $p^s c x_1$ has exponent $r - s$, hence y has exponent $n + r - s$. If $r = s$, then y and $\overline{y}$ have the same exponent. If $s < r$, then $n + r - s \leqslant r \implies n \leqslant s$ since r is exponent of E. Hence

$$y - p^{s-n} c x_1$$

is a lift of $\overline{y}$ with exponent n. Let y_i denote the lift of $\overline{y}_i$ with same exponent.

For the second part of the assertion, suppose there are $a, a_1, \ldots, a_n \in R$ such that

$$ax_1 + a_1y_1 + \cdots + a_ny_n = 0.$$

Going modulo (x_1), we get

$$a_1\overline{y}_1 + \cdots + a_n\overline{y}_n = 0.$$

By hypothesis, $a_i\overline{y}_i = 0$ for all i. Let r_i denote the exponent of $\overline{y}_i$, then $p^{r_i} \mid a_i$. We can hence conclude that $a_iy_i = 0$, finally giving $ax_1 = 0$. □

Remark 1.35. Let E is a finitely generated torsion module. Further, let $E = E(p)$ for some prime $p \in R$. We can write

$$E = \frac{R}{(p^{\nu_1})} \oplus \cdots \oplus \frac{R}{(p^{\nu_s})}$$

where $1 \leqslant \nu_1 \leqslant \cdots \leqslant \nu_s$. Further, these ν_i are unique.

Proof. Let m denote the cardinality of the minimal generating set of E. We induct on m. For $m = 1$, there is some $x_1 \in E$ such that $E = Rx_1$. Consider the map $\varphi : R \to E$ where $r \mapsto rx_1$, this is a surjection. Observe that $\ker \varphi = \mathrm{Ann}_R(x_1) = (p^{\nu_1})$ for some $\nu_1 \geqslant 1$. By the first isomorphism theorem,

$$E \cong \mathrm{Im}(\varphi) \cong \frac{R}{\ker \varphi} \cong \frac{E}{(p^{\nu_1})}. \tag{$*$}$$

For $m > 1$, let ν_1 be the exponent of E. Let $x_1 \in E$ have exponent ν_1. Put $\overline{E} = E/Rx_1$. Note that $x_1 \notin pE$, for if it were then $x_1 = pw$ for some $w \in E$ but $p^{\nu_1 - 1}$ annihilates pw. Let $\pi : E \to E/pE$ denote the canonical projection. By 1.32, $m = \dim_{R/pR} E/pE$, hence there are $z_2, \ldots, z_m \in E$ such that $\pi(x_1), \pi(z_2) \ldots, \pi(z_m) \in E/pE$ are a basis of E/pE are a R/pR-vector space. Using 1.31, we get that $x_1, z_2, \ldots, z_m$ generates E. Further, $\overline{z_2}, \ldots, \overline{z_m} \in \overline{E}$ generate $\overline{E}$. This shows that the minimal generating set of $\overline{E}$ is strictly less than m. Observe that $\mathrm{Ann}_R(\overline{E}) = (p^k)$ for some $k \leqslant \nu_1$. By induction hypothesis, there are independent elements $\overline{y}_2, \ldots, \overline{y}_s \in \overline{E}$ and $\nu_2 \geqslant \cdots \geqslant \nu_s$ such that $\overline{E}$ has the decomposition

$$\overline{E} \cong (\overline{y}_2) \oplus \cdots \oplus (\overline{y}_s) \cong \frac{R}{(p^{\nu_2})} \oplus \cdots \oplus \frac{R}{(p^{\nu_s})}.$$

By 1.34, we get lifts $x_i \in E$ of $\overline{y}_i$ for each $2 \leqslant i \leqslant s$ such that $x_1, \ldots, x_s$ are independent. Let ν_2 be the exponent of $\overline{E}$. By ($*$), we get $(x_1) \cong R/(p^{\nu_1})$. Since $E = (x_1) + E/(x_1)$ (abusing notation), we finally get

$$E = \frac{R}{(p^{\nu_1})} \oplus \frac{R}{(p^{\nu_2})} \oplus \cdots \oplus \frac{R}{(p^{\nu_s})}.$$

Since ν_1 was maximal, we get $\nu_1 \geqslant \nu_2 \geqslant \cdots \geqslant \nu_s$.

We now prove uniqueness. Let $M = R/(ab)$ for some $a, b \in R \setminus \{0\}$. Let $\pi : aR \to aR/(ab)$ be the canonical projection. Observe that $\ker \pi = (a)$ since R. Hence $R/(a) \cong bR/(ab)$.

Let r_i denote the dimension of $p^{i-1}E/p^iE$ as an R/pR vector space. Note that

$$\frac{E}{pE} \cong \frac{R/(p^{\nu_1}) \oplus \cdots \oplus R/(p^{\nu_s})}{pR/(p^{\nu_1}) \oplus \cdots \oplus pR/(p^{\nu_s})} \cong \underbrace{\frac{R}{pR} \oplus \cdots \oplus \frac{R}{pR}}_{s \text{ times}}.$$

Hence $r_1 = s$. Observe that p^kE kills all terms $R/(p^h)$ in the decomposition where $h \leqslant k$. Therefore, all terms with exponent 1 will die in pE, hence r_2 will be the number of terms with exponent greater than or equal to 2. More generalty, $r_n - r_{n-1}$ is the number of terms with exponent equal to $n-1$. Since this is invariant of the decomposition chosen, we get our uniqueness condition. □

Theorem 1.36. *(Structure Theorem) Let R be a PID and let E be a finitely generated R-module. Then there is $m \geqslant 0$ and nonzero $q_1, \ldots, q_r \in R$ where $q_1 \mid q_2 \mid \cdots \mid q_r$ such that*

$$E \cong R^m \oplus \frac{R}{(q_1)} \oplus \frac{R}{(q_2)} \oplus \cdots \oplus \frac{R}{(q_r)}.$$

The sequence of ideals $(q_1), \ldots, (q_r)$ are uniquely determined.

Proof. Using 1.28, we can split E as $E = E/E_{tor} \oplus E_{tor}$ where E/E_{tor} is free. Let $E = E_{tor}$, using 1.34, we have a decomposition of E into p-submodules $E(p_1) \oplus \cdots \oplus E(p_n)$. Using 1.35, we can further decompose each $E(p_i)$ into a direct sum

$$E(p_i) = \bigoplus_{j=1}^{r_i} R/(p_i^{\nu_{ij}}).$$

Let $r = \max_i r_i$. Renumber ν_{ij} such that[2]

$$\nu_{ij} \mapsto \begin{cases} 0 & j \leqslant r - r_i \\ \nu_{i,(r-r_i+j)} & \text{otherwise} \end{cases}$$

We can represent this symbolically as,

$$\begin{array}{cccccccc} E(p_1): & \nu_{11} & \leqslant & \nu_{12} & \leqslant & \cdots & \leqslant & \nu_{1r} \\ E(p_2): & \nu_{21} & \leqslant & \nu_{22} & \leqslant & \cdots & \leqslant & \nu_{2r} \\ \vdots & \vdots & & \vdots & & \ddots & & \vdots \\ E(p_n): & \nu_{n1} & \leqslant & \nu_{n2} & \leqslant & \cdots & \leqslant & \nu_{nr} \end{array}$$

Define $q_i = p_1^{\nu_{1i}} p_2^{\nu_{2i}} \cdots p_n^{\nu_{ni}}$ for each $i = 1 \ldots r$. Observe that $q_1 \mid q_2 \mid \cdots \mid q_r$. Recall that if $a, b \in R$ are coprime then $R/(a) \oplus R/(b) \cong R/(ab)$ as R modules. Hence we get the desired result. The uniqueness follows from 1.35 and 1.33. $\square$

2. Unsorted

Remark 2.1. Let $M_1, \ldots, M_n$ and N be R-modules. We say that the map $\varphi : \prod_i M_i \to N$ is R-multilinear if for all $i = 1, \ldots, n$ and for all $m_i \in M_i$, then map $m_i \mapsto \varphi(m_1, \ldots, m_n)$ is R-linear.

We denote the set of bilinear maps from M to N by $\mathrm{Bi}(M, N)$.[3]

Remark 2.2. Let M, N, P be R-modules and let $\varphi \in \mathrm{Bi}(M \times N, P)$. The tensor product $M \otimes_R N$ is an R-module equipped with a bilinear map $\otimes : M \times N \to M \otimes N$ such that there is a unique R-linear map $\overline{\varphi} : M \otimes N \to P$ such that

$$\begin{tikzcd} M \times N \arrow[r, "\varphi"] \arrow[d, "\otimes"'] & P \\ M \otimes_R N \arrow[ur, "\exists! \overline{\varphi}"'] & \end{tikzcd}$$

Essentially, every bilinear map from $M \times N$ *factors through* $M \otimes_R N$.

Remark 2.3. The uniqueness of such an object follows from the uniqueness of a universal object. To show that such an object exists, we give an explicit construction. Let $E = R^{M \times N}$, the free module on the set $M \times N$ and let $\alpha : M \times N \to E$ be the natural inclusion. Let I be the submodule generated by the elements:

$$\{\alpha(r_1 m_1 + r_2 m_2, n) - r_1\alpha(m_1, n) - r_2\alpha(m_2, n) \mid n \in N; m_1, m_2 \in M; r_1, r_2 \in R\} \cup$$
$$\{\alpha(m, r_1 n_1 + r_2 n_2) - r_1\alpha(m, n_1) - r_2\alpha(m, n_2) \mid m \in M; \quad n_1, n_2 \in N; r_1, r_2 \in R\}.$$

We now verify that $\widetilde{E} = E/(I)$ satisfies the universal properties of tensor products. Let $\varphi \in \mathrm{Bi}(M \times N, P)$. By 1.15, there is a unique R-linear map $\Psi : E \to P$. It follows from the bilinearity of φ that Ψ vanishes on I. Hence, by the universal property of the quotient, there is a unique R-linear map from $\widetilde{E} \to P$. The map $\otimes = \pi \circ \alpha$, as seen below.

$$\begin{tikzcd} M \times N \arrow[r, "\varphi"] \arrow[d, "\alpha"'] & P \\ E \arrow[ur, "\Psi"] \arrow[r, "\pi"'] & \widetilde{E} \arrow[u, "\overline{\Psi}"'] \end{tikzcd}$$

[2]This is just being pedantic, we are essentially reordering to make the maximum ν_{ij} of each i to start from the right side and whatever slots get left over become 0. We define q_i to be the product of each column.

[3]Not to be confused with the set of bimonotonic functions!

Remark 2.4. The elements of $M \otimes_R N$ are called tensors. Elements of the type $x \otimes y$ are called elementary tensors and, generally, a tensor is a R-sum of elementary tensors.

Remark 2.5. If M and N are finitely generated R-modules with generators $\{x_i\}_{i=1}^n$ and $\{y_j\}_{j=1}^m$ respectively, then $M \otimes_R N$ is finitely generated with generators $\{x_i \otimes y_j\}_{i,j}$. Indeed, for an elementary tensor $m \otimes n$, the bilinearity of $\otimes$ gives us

$$m \otimes n = \left(\sum_i a_i x_i\right) \otimes \left(\sum_j b_j y_j\right) = \sum_{i,j} a_i b_j x_i \otimes y_j$$

Remark 2.6. For submodules $I, J \subset R$, there is a unique R-module isomorphism

$$\frac{R}{I} \otimes_R \frac{R}{J} \cong \frac{R}{I+J}$$

where $\overline{x} \otimes \overline{y} \mapsto \overline{xy}$.

Remark 2.7. Let $I \subset R$ be a submodule, then there is a unique R-module isomorphism

$$\frac{R}{I} \otimes_R M \cong \frac{M}{IM}$$

where $\overline{r} \otimes m \mapsto \overline{rm}$.

Remark 2.8. There exist unique isomorphisms such that

1. $M \otimes_R N \cong N \otimes_R M$.
2. $R \otimes_R M \cong M$.
3. $M \otimes_R (N \otimes_R P) \cong (M \otimes_R N) \otimes_R P$.
4. $M \otimes_R (N \oplus P) \cong (M \otimes_R N) \oplus (M \otimes_R P)$. More generally,

$$M \otimes_R \left(\bigoplus_{i \in I} N_i\right) \cong \bigoplus_{i \in I} (M \otimes_R N_i)$$

Remark 2.9. $M_1 \to M_2 \to M_3 \to 0$ is exact if and only if

$$\mathrm{Hom}(M_1, N) \to \mathrm{Hom}(M_2, N) \to \mathrm{Hom}(M_3, N) \to 0$$

is exact for all R-modules N. Similarly, $0 \to M_1 \to M_2 \to M_3$ is exact if and only if

$$0 \to \mathrm{Hom}(N, M_1) \to \mathrm{Hom}(N, M_2) \to \mathrm{Hom}(N, M_3)$$

is exact.

Remark 2.10. Hom-tensor adjunction.

$$\mathrm{Hom}(M \otimes N, P) \cong \mathrm{Hom}(M, \mathrm{Hom}(N, P))$$

The functors $\mathrm{Hom}(N, -)$ and $- \otimes N$ are adjoint.

Remark 2.11. Given module maps $\varphi : M \to M'$ and $\psi : N \to N'$, the map $\varphi \otimes \psi : M \otimes M' \to N \otimes N'$ is defined as $x \otimes y \mapsto \varphi(x) \otimes \psi(y)$. Note that this map is bilinear.

Remark 2.12. The functor $N \otimes -$ is an endofunctor on R-mod where $M \mapsto N \otimes M$ and $\varphi \mapsto id_N \otimes \varphi$. The functor $\mathrm{Hom}_R(N, -)$ (covariant) is an endofunctor on R-mod where $M \mapsto \mathrm{Hom}_R(N, M)$ and $\varphi \mapsto (f \mapsto \varphi \circ f)$. Similarly, the functor $\mathrm{Hom}_R(-, N)$ (contravariant) is an endofunctor on R-mod where $M \mapsto \mathrm{Hom}_R(M, N)$ and $\varphi \mapsto (f \mapsto f \circ \varphi)$.

Remark 2.13. (Restriction and Extension of Scalars) Let $f : R \to S$ be a ring map.

If M is an S module then it can be made into an R module by defining $r \cdot m = f(r) \cdot m$. This is called restriction of scalars. Note that this makes S into an R-module.

On the other hand, let N be an R-module. Since S is an R-module by above, we can take $S \otimes_R N$ which is an R-module. We can define an S-module structure on $S \otimes_R N$ as $s \cdot (s' \otimes n) = (ss') \otimes n$. This is called extension of scalars. Note that the scalars of R can still jump around, this is additional structure added on $S \otimes_R N$.

Remark 2.14. If M is finitely generated over as a module over S and S is finitely generated as a module over R, then M is a finitely generated R module.

Proof. See [Ati94] $\square$

Remark 2.15. If N is finitely generated as a module over R, then $S \otimes_R M$ is finitely generated as an S-module.

Proof. See [Ati94] $\square$

Remark 2.16. Let F and F' be free R-modules with bases $\{e_i\}_{i \in I}$ and $\{f_j\}_{j \in J}$ respectively. Then $F \otimes_R F'$ is free with basis $\{e_i \otimes f_j\}_{i,j}$.

Remark 2.17. An R-algebra is a ring map $\varphi : R \to S$ where $\varphi(R) \subset Z(S)$. This makes S into an R-module and the multiplication on S is compatible with the action of R as

$$(r_1 \cdot s_1)(r_2 \cdot s_2) = f(r_1)s_1 f(r_2)s_2 = f(r_1)f(r_2)s_1s_2 = f(r_1r_2)s_1s_2 = (r_1r_2) \cdot (s_1s_2).$$

Let R-alg denote the category of R-algebras, note that this is a subcategory of R-mod. The morphisms in this category are such that

$$\begin{tikzcd}
S \arrow[rr, "\psi"] & & S' \\
& R \arrow[ul, "\varphi"] \arrow[ur, "\varphi'"'] &
\end{tikzcd}$$

Remark 2.18. (Tensor Algebra) Define $T^n(M)$ as follows

$$T^n(M) = \begin{cases} R & n = 0 \\ M & n = 1 \\ \underbrace{M \otimes \cdots \otimes M}_{m \text{ times}} & \text{otherwise} \end{cases}$$

Further, define

$$T(M) = \bigoplus_{n \geqslant 0} T^n(M)$$

This inherits R-mod structure and addition. Define multiplication $T^n(M) \times T^k(M) \to T^{n+k}(M)$ as

$$(x_1 \otimes \cdots \otimes x_n) \cdot (x_{n+1} \otimes \cdots \otimes x_{n+k}) = x_1 \otimes \cdots \otimes x_{n+k},$$

extend R-linearly.

This is an R-algebra, take the inclusion $R = T^0(M) \hookrightarrow T(M)$.

Remark 2.19. $\mathrm{Hom}_{R-alg}(T(M), A) \cong \mathrm{Hom}_{R-mod}(M, A)$. The functors $T : R - mod \to R - alg$ and (forgetful) $\mathscr{F} : R - alg \to R - mod$ are adjoint.

Remark 2.20. If $M = Rx$ then $T(M) \cong R[x]$.

Remark 2.21. If $M = Rx \otimes Ry$ then $T^k(M) \cong R^{2^n}$

3. Appendix

3.1. Category Theory

References

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